

Transient Freezing Governs Flat-Minimum Selection in Stochastic Gradient Descent

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Stochastic gradient descent (SGD) often finds flatter minima with better generalization. Yet the key question remains unresolved: when is the final basin actually selected during training? Existing explanations mainly emphasize a late-stage or quasi-steady-state bias in a fixed landscape. Here we show that the decisive event occurs earlier, during a transient nonequilibrium regime in which SGD remains mobile across competing valleys before inter-valley motion freezes. Using continuation-training measurements, we find direct evidence of valley hopping during early training and define a freezing time that marks the end of effective inter-valley exploration. We then show that stronger SGD noise promotes flatter minima mainly by delaying this freezing time, not simply by increasing an instantaneous static preference. A minimal two-valley model with anisotropic, landscape-dependent noise reproduces the same sequence of phenomena and explains why fixed-landscape steady-state bias alone cannot account for the observed noise dependence. These results recast flat-minimum selection as a history-dependent nonequilibrium selection problem controlled by the duration of transient mobility.

I. INTRODUCTION

Deep neural networks are trained by moving through a high-dimensional loss landscape, yet the dynamical principle that determines the final solution remains only partially understood [1–4]. One of the most robust observations is that stochastic gradient descent (SGD) tends to find solutions that are flatter and often more generalizable than those obtained by less noisy optimization procedures [5–10]. This has led to the common view that SGD has an implicit bias toward flat minima.

The key question, however, is not only *which* minima are preferred, but *when* that preference is converted into a final basin choice. Many influential explanations focus on late-stage dynamics, for example by analyzing the effective potential induced by anisotropic noise or by studying near-stationary motion inside a fixed or slowly varying valley [11–14]. These approaches capture an important part of SGD, but they are most natural after training has already entered a weak-gradient regime. They say less about the early stage, when gradients are large, barriers are evolving, and the trajectory has not yet committed to a single basin.

This distinction matters because many properties of the final network are established surprisingly early in training [15–19]. The decisive stage of optimization may therefore be strongly nonequilibrium. If so, then the relevant issue is not only which basin would be favored in a hypothetical steady state, but also how long the dynamics remain able to explore competing basins before that exploration arrests.

Here we study flat-minimum selection as a nonequilibrium selection problem controlled by *transient freezing*. We show that SGD initially remains mobile across multiple valleys and that the final outcome is set when effective inter-valley motion ceases. Stronger noise promotes flatter minima mainly by extending the finite-time window over which basin competition remains active, not simply by strengthening an instantaneous static bias. In this sense the final state is selected by the pathway to arrest rather than by static preference alone.

Our argument proceeds in four steps. First, continuation-training experiments reveal direct valley hopping during early training and identify a crossover to basin commitment. Second, this crossover defines a freezing time whose dependence on noise predicts the final basin selected by SGD. Third, we introduce a minimal two-valley model with anisotropic, landscape-dependent noise that reproduces the same sequence of phenomena in a reduced setting. Finally, we use that model to show why fixed-landscape steady-state bias is real but incomplete. The resulting picture shifts the emphasis from asymptotic sampling to the finite-time arrest of exploration.

II. EARLY-TIME VALLEY HOPPING PRECEDES BASIN COMMITMENT

The most direct evidence for transient selection is that early SGD trajectories are not immediately confined to a single basin. Instead, training remains mobile across multiple valleys before eventually becoming committed to one of them. Two observations in our data support this picture. First, independent runs from the same initialization converge to distinct clusters in parameter space that are separated by substantial interpolation barriers, indi-

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cating the presence of competing valleys. Second, and more directly, continuation-training experiments show that the valley associated with a partially trained network is still changing during the early stage of optimization.

The continuation protocol is simple. A reference SGD trajectory is saved at a sequence of checkpoint times t_c . From each checkpoint, training is restarted using deterministic full-batch gradient descent. Because the continuation dynamics contain little or no stochastic valley hopping, the final state of each continuation identifies the valley that the SGD trajectory would occupy if exploration were arrested at time t_c . For early checkpoints, different continuation times lead to final states with substantially different flatness and test performance. For sufficiently late checkpoints, all continuations converge to the same valley. The crossover between these two regimes shows that early training is a period of genuine basin competition.

This observation suggests an operational definition of basin commitment. Let $v(t_c)$ denote the valley identity inferred from a continuation started at checkpoint t_c . Early in training, $v(t_c)$ depends strongly on t_c ; later, it becomes stable. We therefore define the freezing time t_f as the earliest time after which the final valley identity remains robust under continuation or independent reruns. Conceptually, t_f is not simply the time at which the loss becomes small. It is the time at which effective inter-valley rearrangements cease.

The important point is that the decisive event in SGD is not the final relaxation inside a chosen valley. It is the loss of mobility across valleys. Once that mobility is gone, later optimization mainly refines an already selected solution. This makes the early nonequilibrium regime the natural place to look for the origin of flat-minimum selection.

III. BASIN SELECTION IS SET AT THE FREEZING POINT

Once basin commitment is identified as the decisive event, the next question is when that event occurs. Our central empirical claim is that the relevant organizing variable is the duration of the transient exploratory window, quantified by the freezing time t_f . In practice, the continuation-based definition above can be complemented by a proxy derived directly from the training dynamics: the time at which the trajectory enters a low-loss, effectively committed regime. In the current dataset and architecture, this proxy is well captured by

$$t_f \equiv \min\{t \mid \mathcal{L}_{\text{train}}(t) < \mathcal{L}_c\}, \quad (1)$$

where $\mathcal{L}_c$ is a small threshold chosen to mark the end of the transient search phase.

This definition is meaningful because the continuation experiments show that the same crossover also marks the disappearance of effective valley hopping. Runs with

larger SGD noise—achieved, for example, by increasing the learning rate or decreasing the batch size—reach this committed regime later. Their training trajectories remain mobile across competing valleys for longer, and this prolonged mobility increases the probability of ending in the flatter valley.

The key point is that the relevant parameter is not simply the instantaneous amplitude of the noise. What matters is when inter-valley mobility is lost. Two optimization settings with different microscopic noise statistics can produce similar outcomes if they generate similar freezing times, whereas settings with comparable nominal noise amplitudes can produce different outcomes if one of them freezes much earlier. Freezing time therefore provides a more direct organizing variable for basin selection than raw hyperparameters alone.

Empirically, later freezing correlates with both larger flatness and improved generalization in our experiments. The transient noisy phase is thus not wasted wandering. It is the finite-time search process through which SGD discovers more favorable valleys before it loses the ability to move between them. The final outcome is set at the freezing point, not by t_f in isolation, but by the landscape state and valley bias that remain when exploration stops.

IV. A MINIMAL THEORY OF TRANSIENT FREEZING

To identify the minimal ingredients required for transient selection, we consider a reduced two-dimensional landscape $\mathcal{L}(x, y)$. The coordinate y represents the downhill training direction, whereas x resolves competition between two valleys of unequal flatness. As training proceeds, the landscape bifurcates and produces a flatter valley for $x > 0$ and a sharper valley for $x < 0$. A convenient explicit realization is

$$\mathcal{L}(x, y) = \mathcal{L}_0(y) + \begin{cases} \frac{x[x - g_1(y)]}{f_1(y)}, & x \geq 0, \\ \frac{x[x + g_2(y)]}{f_2(y)}, & x < 0, \end{cases} \quad (2)$$

where $g_{1,2}(y)$ control the valley positions and $f_{1,2}(y)$ their local flatness. For clarity we denote $f_{\text{flat}}(y) \equiv f_1(y)$ and $f_{\text{sharp}}(y) \equiv f_2(y)$, with flatness contrast $\gamma \equiv f_{\text{flat}}/f_{\text{sharp}} > 1$. The barrier height separating the two minima at fixed y is denoted by $\Delta\mathcal{L}(y)$. As y increases, the valleys separate, the barrier grows, and both valleys flatten. These are the minimal ingredients suggested by the neural-network experiments.

The stochastic dynamics are modeled by a discrete-time Langevin equation,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta [\nabla \mathcal{L}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_t], \quad \boldsymbol{\theta} = (x, y), \quad (3)$$

with a noise covariance chosen to mimic the anisotropic, landscape-dependent structure of SGD,

$$\text{Cov}(\boldsymbol{\xi}_t) = 2\sigma \mathbf{H}(\boldsymbol{\theta}_t), \quad (4)$$

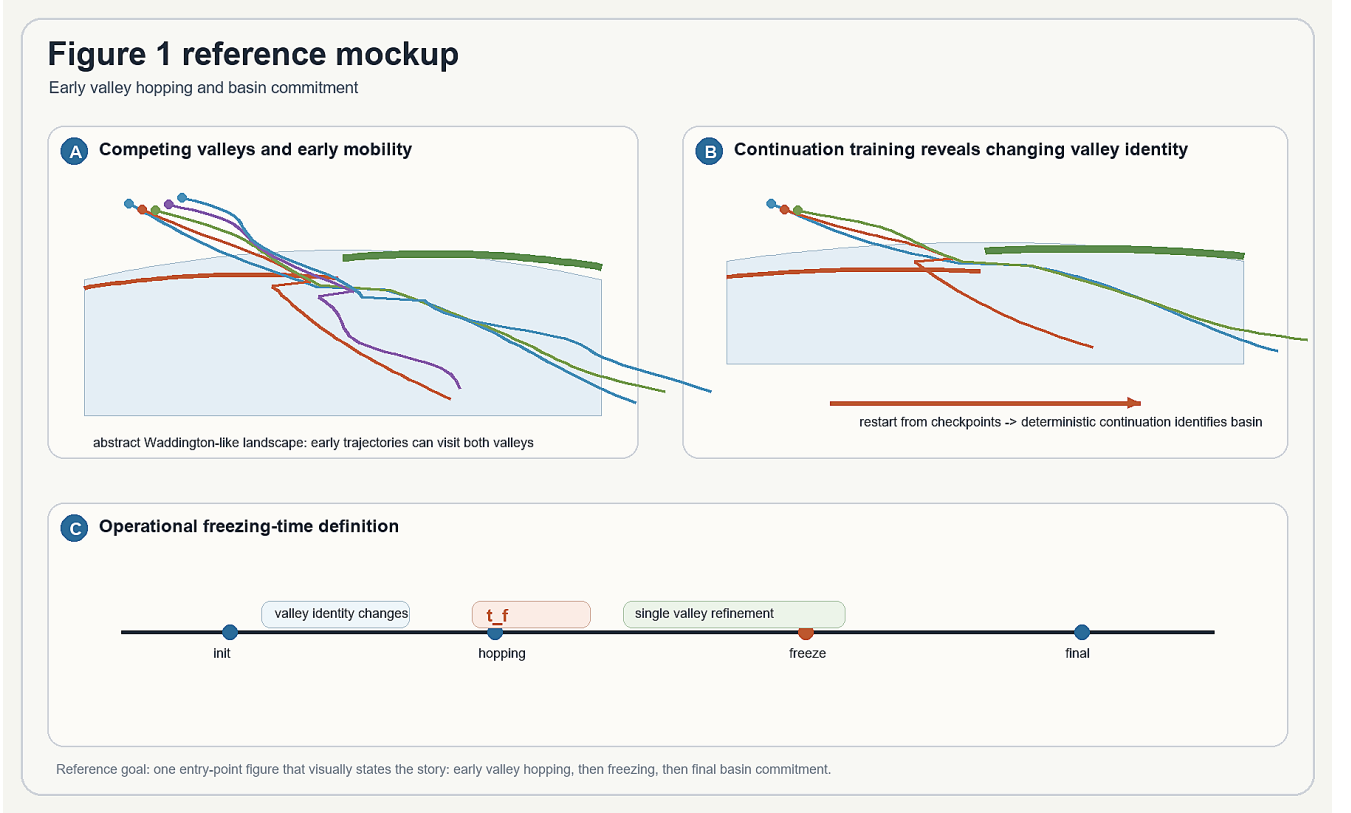


FIG. 1. Early valley hopping precedes basin commitment. This rough composite reuses the most relevant panels from the previous draft to provide a first version of the new entry-point figure. The upper-left panel summarizes the existence of multiple solution valleys in parameter space. The upper-right panel shows continuation-training trajectories, which reveal that the valley associated with a partially trained network is still changing during early training. The bottom panel shows the training-loss/flatness dynamics used to define a practical freezing-time proxy. Together these panels motivate the central claim of the present draft: SGD initially remains mobile across competing valleys, and the final basin is selected when this inter-valley mobility freezes. In the final figure version, these ingredients can be redrawn more compactly around a single operational definition of t_f .

where $\mathbf{H}$ is the local Hessian and σ sets the noise scale. The effective noise level is therefore $\Delta_S \sim \eta\sigma$. In this reduced description, Δ_S controls the exploratory mobility, $\Delta\mathcal{L}(y)$ sets the barrier to valley exchange, and γ measures the geometric asymmetry between the two basins.

Despite its simplicity, the model reproduces the sequence emphasized by the data. Early in the dynamics, when barriers are still low, trajectories hop repeatedly between the two valleys. As y increases, the barrier grows and valley-to-valley transitions become progressively rarer. The final basin is selected when this hopping effectively stops. Increasing Δ_S keeps the system mobile for longer, thereby postponing the freezing point and increasing the probability of ending in the flatter valley.

The model is not intended as a literal reduction of deep learning to two coordinates. Its purpose is to isolate the minimal mechanism: competing valleys, geometry-coupled noise, a growing barrier, and a finite exploratory window terminated by arrest. Once these ingredients are present, flat-minimum selection becomes

a history-dependent consequence of nonequilibrium dynamics rather than merely an asymptotic property of a fixed landscape.

V. STATIC BIAS IS REAL BUT INSUFFICIENT

With the reduced model in hand, we can now ask what a fixed-landscape picture predicts. In a quasi-steady regime where the fast inter-valley coordinate relaxes at an approximately fixed training coordinate y , anisotropic SGD noise induces an effective nonequilibrium bias toward flatter valleys [11, 12, 14]. In the present model, that quasi-steady preference can be written as the occupation probability of the flat valley,

$$P_{\text{flat}}^{\text{ss}}(y) \approx \left[1 + \gamma^{-\frac{1}{2}} \exp\left(\frac{\Delta\mathcal{L}(y)[f_{\text{sharp}}(y) - f_{\text{flat}}(y)]}{2\Delta_S}\right) \right]^{-1}, \quad (5)$$

where $\gamma = f_{\text{flat}}/f_{\text{sharp}} > 1$ is the flatness contrast. This relation captures a real fixed-slice bias: geometry-coupled



FIG. 2. Basin selection is set at the freezing point. This temporary figure combines the previous early-dynamics and solution-properties figures. The upper part shows that stronger SGD noise delays entry into the effectively frozen low-loss regime, thereby increasing the freezing time t_f . The lower part summarizes the corresponding change in final outcomes across noise conditions: broader exploration is associated with flatter and better-generalizing solutions. The intended message is that raw noise strength is not the most direct organizing variable; rather, noise affects the final solution mainly by shifting when basin competition is frozen into the dynamics.



FIG. 3. Minimal two-valley model for transient freezing. This provisional figure reuses the toy-model figure from the previous manuscript. It introduces the bifurcating landscape with a flatter and a sharper valley, together with anisotropic, landscape-dependent noise. The simulation panels show the two key observables required in the present draft: the probability of ending in the flat valley and the freezing time, both as functions of effective noise. The figure therefore already captures the minimal mechanism emphasized in the new outline: competing valleys, evolving barriers, and delayed freezing under stronger noise.

noise favors the broader valley.

The same relation also shows why static bias is not the whole story. At fixed y , increasing the effective noise level Δ_S does not necessarily make the flatter valley more selectively favored. Once the landscape slice is fixed, larger noise can reduce instantaneous selectivity by making the



FIG. 4. Why static bias is insufficient. This rough version is based on the analytical figure from the previous draft. It already contains the key ingredients needed here: anisotropic SGD noise induces a quasi-steady preference for flatter valleys in an effective landscape, but the final selection law is set only after the dynamics freeze. The figure therefore supports the conceptual distinction emphasized in the present manuscript between a fixed-landscape instantaneous bias and the actual training outcome, which depends on when inter-valley motion becomes negligible. In the final version, this figure should be simplified so that the paradox and its resolution are visually more explicit.

occupation probability more mixed. This is in tension with the empirical trend that stronger SGD noise often leads to *more* convergence to flatter minima, not less.

The missing ingredient is that the landscape evolves while inter-valley motion shuts down. The final training outcome is not determined by a steady-state probability evaluated at an arbitrary fixed position. It is determined by the quasi-steady bias that is inherited when hopping arrests. Stronger noise can therefore weaken instantaneous selectivity at fixed y while still enhancing the final selection of the flatter valley, because it delays the time at which the dynamics stop exploring competing basins.

VI. QUANTITATIVE PREDICTIONS AND EXPERIMENTAL TESTS

The transient-freezing framework becomes predictive once we distinguish between two probabilities: the instantaneous quasi-steady preference at a given y , and the final occupation probability selected when exploration freezes. In the two-valley model, the latter is obtained by evaluating the quasi-steady bias at the freezing point,

$$P_{\text{flat}}^{\text{tr}} \approx P_{\text{flat}}^{\text{ss}}(y_f), \quad (6)$$

where y_f is the training coordinate at which inter-valley hopping becomes negligible. Because y_f increases with the effective noise level, the final occupation probability can grow with noise even when the fixed- y selectivity in Eq. (5) becomes weaker. This is the minimal statement of transient selection.

The theory makes several concrete predictions. First, the freezing time should increase systematically with stronger SGD noise. Second, final flat-valley selection should correlate more directly with freezing time than with raw hyperparameters viewed in isolation. Third, a minimal model and a neural network need not share microscopic details to exhibit the same selection law, provided they share the same ordering of time scales: rapid equilibration across valleys early on, followed by a barrier-driven arrest of inter-valley motion.

Our existing experiments already support each part of this picture at a qualitative level. Noise delays the end of the transient search phase; continuation training shows that valley identity remains labile precisely during that window; and flatter, better-generalizing solutions become more likely when that window is extended. The minimal model reproduces the same trends and explains why static bias and final selection need not have the same dependence on noise.

A particularly direct test in the final version will be to organize the data by t_f itself. If transient freezing is the correct organizing principle, then different combinations of batch size and learning rate should become more comparable once they are viewed through the freezing-time lens. Whether or not the final data collapse is perfect, the theory identifies t_f as the physically meaningful variable connecting stochasticity to the geometry of the selected solution.

VII. DISCUSSION: FLAT-MINIMUM SELECTION AS A NONEQUILIBRIUM FREEZING PROCESS

In this work, we argue that flat-minimum selection in SGD is best viewed as a nonequilibrium freezing problem. The central mechanism is simple: SGD initially explores competing valleys, and the final basin is selected when effective inter-valley motion stops. The relevant question is therefore not only which valley is favored in a notional steady state, but when basin competition freezes.

This perspective clarifies the role of SGD noise. Noise does not simply act as a scalar regularizer that favors flatness at all times. Its main role is dynamical: it controls the duration of the exploratory regime before freezing. Stronger noise can therefore enhance flat-minimum se-

lection even when fixed-landscape selectivity alone would not predict such an enhancement. This suggests a practical design principle for optimization schedules: to influence the final solution, control not only the asymptotic noise level but also the time at which basin commitment occurs.

Our analysis also suggests why the mechanism should extend beyond the specific network and dataset studied here. The required ingredients are generic: multiple competing valleys, geometry-coupled stochasticity, barriers that evolve during descent, and a finite window of exploration. Whenever these ingredients coexist, one should expect a transient-freezing problem rather than a purely steady-state sampling problem. The framework should therefore be relevant to deeper networks, richer datasets, and more general stochastic optimizers.

Several directions remain for the final paper. The quantitative theory should be tightened, the strongest architecture-level evidence should be moved into the main text, and the final figures should be rebuilt around the transient-freezing narrative. More generally, it will be important to determine how broadly freezing time can organize optimization outcomes across architectures and training protocols. But the central message is already clear: SGD selects flat minima not simply because they are statically favored, but because noise keeps the dynamics mobile across competing valleys until a sufficiently late freezing time allows the flatter valley to win.

DATA AND CODE AVAILABILITY

The code used for the empirical experiments and the two-valley simulations is available at https://github.com/YangNing1995/Transient_Dynamics_SGD.

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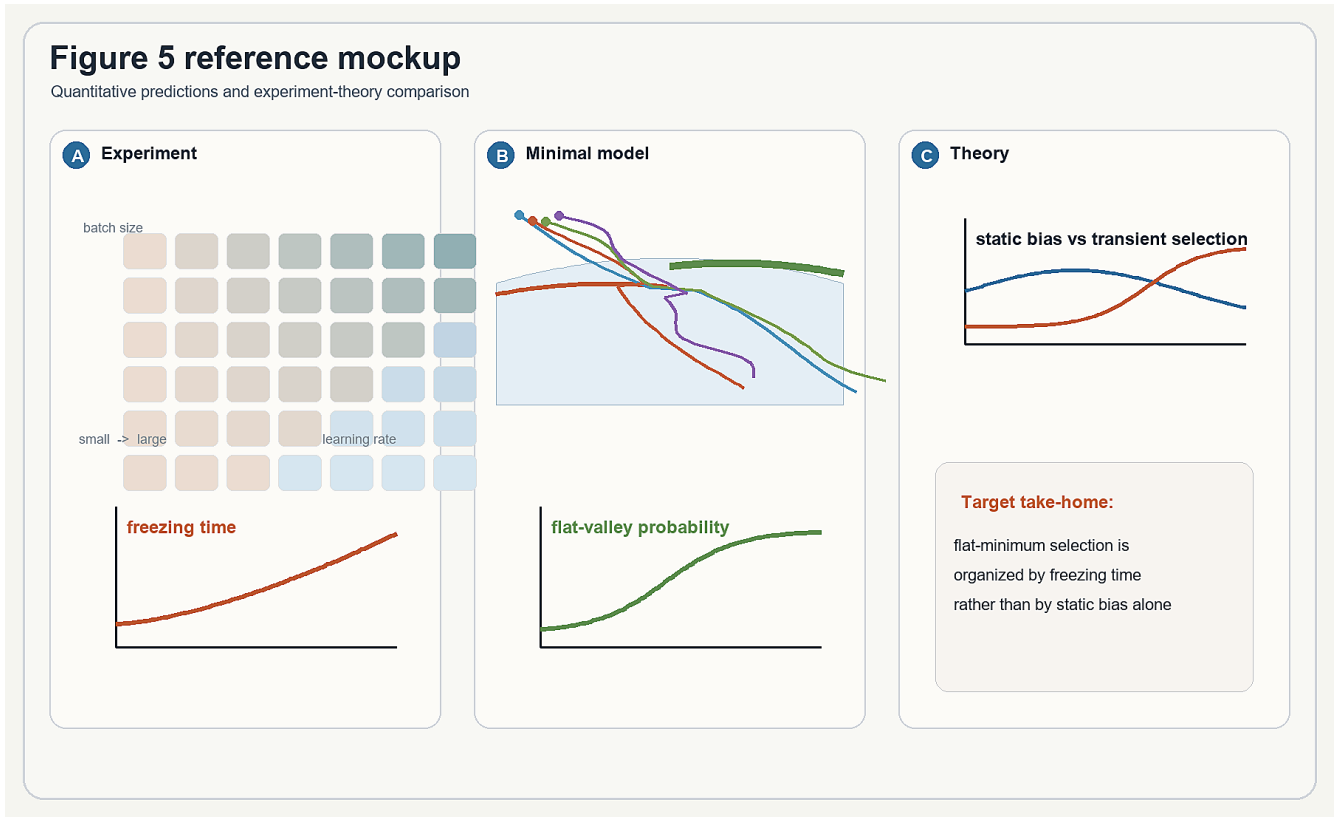


FIG. 5. Quantitative predictions and experiment-theory comparison. This temporary summary figure assembles three ingredients from the previous draft: neural-network evidence for noise-dependent flatness/generalization, the minimal two-valley model, and the analytical transient-selection theory. As a rough version, it already supports the structure of the revised paper: experiments show that stronger noise correlates with flatter solutions, the toy model reproduces hopping and delayed freezing, and the analytical theory explains why the final flat-valley occupation is controlled by the freezing point rather than by static bias alone. In the final PRX figure set, this composite can be replaced by a cleaner side-by-side comparison organized explicitly by t_f .

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